

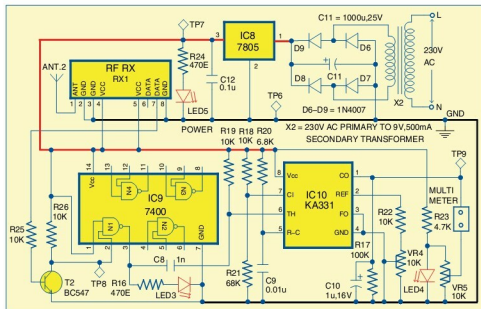


Fig. 3: Circuit of the receiver unit

built around operational amplifier IC2. The output signal from pin 6 of IC2 is further fed to V-F converter section comprising an integrator wired around operational amplifier IC3 and V-F converter IC4. Integrator IC3 improves the

V-F converter's linearity in conversion process.

The frequency-encoded temperature data is then sent to transmitter module TX1 via opto-coupler IC5 and buffer transistor T1. Flickering of LED2 indicates ongoing V-F conversion.

To derive power supply for the circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 9V-0-9V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitors C1 and C2, and regulated by ICs 7805 and 7905 (IC6 and IC7) for +5V and -5V, respectively. Capacitors C3 and C4 bypass ripples present in the regulated supply.

Fig. 3 shows the receiver circuit. It comprises NAND gate IC 7400 (IC9), F-V converter KA331 (IC10), regulator 7805 (IC8), 433MHz ASK receiver module (RX1) and a few discrete components. RX1 is used to receive and demodulate the ASK-modulated RF signal transmitted from the transmitter unit. The demodulated output is a train of rectangular pulses as already explained in the transmitter section.

Transistor T2 is used to amplify and at the same time limit the amplitude of the pulse between 0 and 5 V. This TTL-compatible output is then fed to IC9.

The NAND gate helps to get perfectly rectangular wave shaped pulses. LED3 at pin 3 of IC9 flickers to indicate reception of the demodulated signal. The output of IC9 is also fed to F-V converter IC10 through capacitor C8. It generates a voltage equivalent to the frequency of the demodulated signal from receiver module RX1.

To get the actual sensed signal voltage developed by sensor LM35, the voltage output at pin 1 of IC10 has to be reduced by a voltage equal to the reference voltage developed by LED1 of the transmitter unit. In order to do this, a stable voltage is first developed across LED4.

The required reference voltage is then achieved by adjusting preset VR5. Preset VR5 is pre-adjusted during calibration to generate this reference voltage.

To derive power supply for the receiver circuit, the 230V AC mains is stepped down by transformer X2 to deliver a secondary output of 9 V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D6 through D9, filtered by capacitor C11 and regulated by IC 7805 (IC8). Capacitor C12 bypasses ripples present in the regulated supply. LED5 acts as the power indicator and R24 limits the current through LED5.

Construction

An actual-size, single-side PCB for the transmitter circuit (Fig. 2) is shown in Fig. 4 and its component layout in Fig. 5. PCB for the receiver circuit (Fig. 3) is shown in Fig. 6 and its component layout in Fig. 7. Assemble the circuits on PCBs to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC base and, before inserting the IC, check the supply voltage. Refer test points given in the table to check the circuit for proper functioning.

For proper operation of the remote sensing thermometer, pre-adjustment

PARTS LIST

Semiconductors:	
IC1	- LM35 temperature sensor
IC2, IC3	- CA3140 operational amplifier
IC4, IC10	- KA331 voltage-to-frequency converter
IC5	- MC2TE opto-coupler
IC6, IC8	- 7805, 5V regulator
IC7	- 7905 -5V regulator
IC9	- 7400 NAND gate
T1, T2	- BC547 npn transistor
D1-D9	- 1N4007 rectifier diode
LED1-LED5	- 3mm LED
TX1	- 433MHz ASK transmitter module
RX1	- 433MHz ASK receiver module
Resistors (all 1/4-watt, 45% carbon):	
R1	- 91-kilo-ohm
R2-R4	- 100-kilo-ohm
R5	- 2.2-kilo-ohm
R6, R7, R9, R15, R18, R19, R22, R25, R26	- 10-kilo-ohm
R8, R20	- 6.8-kilo-ohm
R10, R14	- 1-kilo-ohm
R11, R12, R13	- 3.3-kilo-ohm
R16, R24	- 470-ohm
R17	- 100-kilo-ohm
R21	- 68-kilo-ohm
R23	- 4.7-kilo-ohm
VR1, VR2, VR4, VR5	- 10-kilo-ohm trim potmeter
VR3	- 20-kilo-ohm trim potmeter
Capacitors:	
C1, C2, C11	- 1000µF, 25V electrolytic
C3, C4, C12	- 0.1µF ceramic disk
C5, C7, C9	- 0.01µF ceramic disk
C6, C10	- 1µF, 16V electrolytic
C8	- 1nF ceramic disk
Miscellaneous:	
X1	- 230V AC primary to 9V-0-9V, 500mA secondary transformer
X2	- 230V AC primary to 9V, 500mA secondary transformer